WEEK 3 · LESSON 18 OF 24 · HANDOUT

Week 3 Project: Statistics of a Class

Study time: **about 60 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=R4yfNi_8Kqw — open the lesson online to watch inside the course.

Your goal in this lesson: Apply vectors, descriptive statistics, distributions and probability to a realistic dataset, and explain the results in plain language.

The dataset

A school wants to understand the results of 120 Senior 6 students in three subjects. The data is **simulated** for practice. Run this cell:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

rng = np.random.default_rng(2026)
n = 120
ability = rng.normal(0, 1, n)
results = pd.DataFrame({
    "maths": (62 + 12 * ability + rng.normal(0, 8, n)).clip(0, 100).round(),
    "physics": (58 + 11 * ability + rng.normal(0, 9, n)).clip(0, 100).round(),
    "english": (66 + 4 * ability + rng.normal(0, 10, n)).clip(0, 100).round(),
    "school": rng.choice(["Day", "Boarding"], size=n, p=[0.6, 0.4]),
})
results["average"] = results[["maths", "physics", "english"]].mean(axis=1).round(1)
print(results.head())
```

Part 1 — Describe (Lesson 15)

For each subject, report the mean, median, SD and IQR. Which subject is hardest? Which has the most varied results? Write two sentences.

Part 2 — Shape (Lesson 17)

Draw a histogram for each subject on one figure. Do they look roughly normal? Mark the mean on each with `axvline`.

Part 3 — z-scores (Lesson 17)

Add a column with each student's maths z-score. List the students with |z| > 2. Are any marks suspicious enough to double-check?

Part 4 — Probability (Lesson 16)

Using the data as your "population": 1. What is P(average ≥ 70)? 2. What is P(maths ≥ 70 | boarding) and P(maths ≥ 70 | day)? Is there a difference? 3. What is P(maths ≥ 70 and english ≥ 70)? Compare it with P(maths ≥ 70) × P(english ≥

70). Are the two subjects independent?

Part 5 — Vectors (Lesson 13)

Treat each student as a vector `[maths, physics, english]`. 1. Find the two students who are most similar (smallest distance). Hint: loop over pairs, or use the code below. 2. Which student is furthest from the class average vector?

```
from itertools import combinations
X = results[["maths", "physics", "english"]].to_numpy()
best = min(combinations(range(n), 2), key=lambda p: np.linalg.norm(X[p[0]] - X[p[1]]))
print(best, X[best[0]], X[best[1]])
```

Part 6 — Report

Write a one-page note for the head teacher: - three findings with numbers, - one chart, - one honest limitation (for example: the data shows a link between boarding and maths marks, but not *why*).

Remember this: Statistics turn 360 marks into a few clear, honest sentences. Centre, spread, shape, probability and similarity are the tools you will use in every AI project.

Name: _____ Date: _____

Activity

Task 1

Complete Parts 1–6 in a Colab notebook. Swap reports with a classmate and check one calculation from their work.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Which statistic tells you which subject had the most varied results?
   - A. The number of students
   - B. The mode of school
   - C. The mean
   - D. The standard deviation (or IQR)
2. A student's maths z-score is 2.4. What does it mean?
   - A. The mark is 2.4
   - B. The mark is below average
   - C. The mark is 2.4 standard deviations above the mean — unusually high
   - D. The data is wrong
3. If P(maths ≥ 70 and english ≥ 70) is much larger than P(maths ≥ 70) × P(english ≥ 70), what does that suggest?
   - A. The subjects are independent
   - B. Probability does not apply
   - C. Good results in one subject go with good results in the other (not independent)
   - D. The data is empty
4. How do you measure how similar two students are across 3 subjects?
   - A. Add their marks
   - B. Compare their names
   - C. Count their subjects
   - D. Compute the distance between their score vectors
5. Which is an honest limitation to mention?
   - A. The data shows a link, but not the reason for it
   - B. There are no limitations
   - C. The results prove boarding causes better marks
   - D. Statistics are always perfect
6. Which chart best shows whether a subject's marks look roughly normal?
   - A. Line chart over time
   - B. Histogram
   - C. Pie chart
   - D. Table

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